

Exercise (2 - A)**(1) Solve the following nonhomogeneous systems of equations;****a) Using the inverse matrix method,****b) Using Gauss-Jordan elimination method**

$$\begin{aligned} \text{(i)} \quad 3x_1 + 6x_2 &= -3 \\ 5x_1 + 7x_2 &= 10 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad x_1 + 2x_2 &= 4 \\ x_1 - x_2 &= 1 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad x_1 + 2x_2 &= -13 \\ 3x_1 - 2x_2 &= 1 \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad x_1 + 2x_2 &= 3 \\ 3x_1 + 6x_2 &= 5 \end{aligned}$$

$$\begin{aligned} \text{(v)} \quad 3x_1 - 2x_2 &= 4 \\ 6x_1 - 4x_2 &= 5 \end{aligned}$$

$$\begin{aligned} \text{(vi)} \quad x_1 + 2x_2 &= 4 \\ 3x_1 + 6x_2 &= 12 \end{aligned}$$

(2) For each of the following systems $AX = B$, find the value of k such that: the system has one solution or infinitely many solutions or no solution.

$$\text{(a)} \quad \begin{bmatrix} 1 & k \\ k & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$

$$\text{(b)} \quad \begin{bmatrix} 1 & 2 \\ k & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix}$$

(3) Solve the following nonhomogeneous systems of equations;**a) Using the inverse matrix method,****b) Using Gauss-Jordan elimination method**

$$\begin{aligned} \text{(i)} \quad -2x_1 + 3x_2 - x_3 &= 1 \\ x_1 + 2x_2 - x_3 &= 4 \\ -2x_1 - x_2 + x_3 &= -3 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad x_1 + 2x_2 + 3x_3 &= 6 \\ 2x_1 + 7x_2 + x_3 &= 13 \\ x_1 + 4x_2 - 6x_3 &= 1 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad x_1 + 2x_2 + 3x_3 &= 6 \\ 3x_1 + x_2 - x_3 &= -2 \\ 2x_1 - 3x_2 + 2x_3 &= 14 \end{aligned}$$

$$[\text{Ans: } x_1 = 2, \quad x_2 = 3, \quad x_3 = 4]$$

$$[\text{Ans: } x_1 = -1, \quad x_2 = 2, \quad x_3 = 1]$$

$$[\text{Ans: } x_1 = 1, \quad x_2 = -2, \quad x_3 = 3]$$

$$\begin{aligned} \text{(iv)} \quad x_1 - x_2 + x_3 &= -1 \\ 2x_1 + x_2 - 3x_3 &= 8 \\ x_1 - 2x_2 + 3x_3 &= -5 \end{aligned}$$

$$\begin{aligned} \text{(v)} \quad x_1 + x_2 + 3x_3 &= 9 \\ 2x_1 - x_2 + x_3 &= 8 \\ -x_1 + 3x_2 + 2x_3 &= 1 \end{aligned}$$

$$[\text{Ans: } x_1 = 1, \quad x_2 = 0, \quad x_3 = -2]$$

$$[\text{Ans: } x_1 = 3, \quad x_2 = 0, \quad x_3 = 2]$$

(4) Solve the following nonhomogeneous systems of equations

$$\begin{aligned} \text{(i)} \quad x_1 + x_2 - 3x_3 &= 4 \\ 2x_1 + x_2 - x_3 &= 2 \\ 3x_1 + 2x_2 + 2x_3 &= 6 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad x_1 + x_2 - x_3 &= 1 \\ 2x_1 + 3x_2 + 2x_3 &= 3 \\ x_1 + 2x_2 + 3x_3 &= 2 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad 2x_1 + x_2 - x_3 + x_4 &= 5 \\ x_2 + x_3 - x_4 &= 3 \\ 3x_3 + 6x_4 &= 6 \end{aligned}$$

$$[x_1 = -2 - 2r, \quad x_2 = 6 + 5r, \quad x_3 = r], [x_1 = 5r, \quad x_2 = 1 - 4r, \quad x_3 = r], [x_1 = 3 - 3r, \quad x_2 = 1 + 3r, \quad x_3 = 2 - 2r, \quad x_4 = r]$$

(5) Solve the following nonhomogeneous systems of equations:

$$\begin{aligned} \text{(i)} \quad x_1 + 2x_2 + 3x_3 &= 9 \\ 2x_1 - x_2 + x_3 &= 8 \\ -x_1 + 3x_2 + 2x_3 &= 4 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad x_1 - 4x_2 + 7x_3 &= 14 \\ 3x_1 + 8x_2 - 2x_3 &= 13 \\ 7x_1 - 8x_2 + 26x_3 &= 5 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad 2x + 3y + 19z - 4w &= 2 \\ x + 2y + 12z - 3w &= 1 \\ -x + 2y + 8z - 5w &= 1 \end{aligned}$$

$$[\text{Ans (i, ii, iii): The system has no solution}]$$

(6) Calculate LU factorization for each of the following matrices;

$$A = \begin{bmatrix} 2 & 1 \\ -4 & -6 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ 8 & 4 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 & -4 \\ 2 & 1 & -2 \\ 6 & 3 & -11 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 8 & 5 \\ 1 & 11 & 4 \end{bmatrix}$$

(7) Using the answers obtained in (6), solve the following systems of equations;

$$\begin{bmatrix} 2 & 1 \\ -4 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 \\ 8 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 & -4 \\ 2 & 1 & -2 \\ 6 & 3 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \begin{bmatrix} 1 & 3 & 2 \\ 2 & 8 & 5 \\ 1 & 11 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

(8) Let $A = \begin{bmatrix} 1 & 6 & 2 \\ 2 & 12 & 5 \\ -1 & -3 & -1 \end{bmatrix}$

- i) Show that A does not have an LU factorization.
- ii) Re-order the rows of A and find an LU factorization of the new matrix.
- iii) Hence solve,

$$\begin{aligned} x_1 + 6x_2 + 2x_3 &= 9 \\ 2x_1 + 12x_2 + 5x_3 &= -4 \\ -x_1 - 3x_2 - x_3 &= 17 \end{aligned}$$
